



Estimation of the value of an inspection and maintenance program: A Bayesian gamma process model

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ABSTRACT

This paper presents a model to quantify the economic value gained by implementation of an inspection and preventive maintenance program for managing an ageing component population. The proposed approach is a refinement of the Bayesian value of information analysis through the consideration of an intricate interaction between parameter and temporal uncertainties associated with the gamma process of degradation. This paper presents an analytical formulation and computational approach to solve this complex problem in a multivariate setting. The paper shows that the economic value is significantly sensitive to the prior information and relative costs of preventive and corrective maintenance. Since the value of inspection is dominated by a reduction in the parameter uncertainty of the gamma process model, lifecycle cost optimization which ignores this aspect would lead to a sub-optimal solution of the problem.

1. Introduction

1.1. Background

Inspection and preventive maintenance programs play a major role in assuring the safety and operability over the life cycle of an engineering system. Inspections are intended to collect information about the state of existing degradation mechanisms affecting components, as well as information about the potential onset of new degradation processes. In this regard, aged industrial facilities that are approaching near the end of life face an unusual decision problem.

The decision problem considered in this paper is concerned with the ageing management of a population of statistically identical components that are exposed to a degradation process. When a plant is near its end of life, a plant operator would be typically reluctant to develop and implement new inspection and preventive replacement (I-PR) programs in order to minimize unnecessary operating costs. A case in point is the corrosion management in the piping system of a nuclear plant that is expected to be shut down in the next 10 year period. The reason is that intrusive inspections of wall thickness of primary piping components in a radioactive environment tend to be tedious with an adverse effect on the safety of inspectors.

Therefore, if the operator believes that degradation is not aggressive, a “run to failure” strategy can be adopted in which only failed components are replaced as failures occur over time. However, this

strategy would be costly if degradation process is so aggressive that it would result in a large number of disruptive failures. A more balanced approach can be devised by collecting information through inspections about the state of degradation, and if justified, taking preventive actions to reduce the number of failures, i.e., economic losses, until the end of life of the plant. Another simple, but drastic, option is to replace all aged components in a preventive manner and reduce the risk significantly.

The plant operator's dilemma is whether or not it is worth developing and implementing an Inspection and Preventive Replacement (I-PR) program for the ageing management. To answer this question, the paper presents a Bayesian pre-posterior formulation of the problem that involves the stochastic gamma process as a model of degradation.

1.2. Problem definition

Consider an engineering infrastructure consisting of a population of n statistically identical and independently operating components of current age of t_1 years. Since excessive degradation leading to a failure will disrupt plant's operation, the operator is interested in implementing an Inspections and Preventive Replacement (I-PR) program to ensure a safe operation over a time interval from t_1 to t_2 , where $t_2 > t_1$.

Since a component's inspection is a tedious, time consuming and costly process, the plant operator has an option of replacing all the

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components as a precautionary measure and avoid the inspection cost. Another option is to continue using components with sufficient resources in hand to make emergency replacements, if component failures occur between t_1 and t_2 .

As mentioned earlier, the plant operator is unsure about the need for an I-PR program, because it depends on the severity of degradation process. In case of a rapidly progressing degradation, an I-PR program would reduce the number of failures and associated losses to the plant. In contrast, an I-PR program would have limited value when the degradation is benign, since a corrective replacement on failure would be a simple and effective strategy to adopt in this case.

This paper presents a probabilistic solution of this decision problem by modelling the ageing related degradation in each component as a stochastic gamma process with uncertain parameters, which reflect a lack of knowledge about the process. This parameter uncertainty is modelled using a suitable prior distributions within the Bayesian framework.

1.3. Conceptual approach

Fig. 1 illustrates key steps of the decision-making problem related to inspection and replacement of a component. The decision tree begins with two branches:

$$e(t_1) = \begin{cases} e_1 : & \text{Inspect} \\ e_2 : & \text{Do not inspect (base case)} \end{cases} \quad (1)$$

Under each option, two actions are possible:

$$a(t_1) = \begin{cases} a_1 : & \text{Replace} \\ a_2 : & \text{Do not replace} \end{cases} \quad (2)$$

The bottom “no-inspection” branch of the decision tree involves a prior analysis that uses prior distributions of gamma process parameters in the computation of expected cost of two actions: replace, or do not replace. The action that minimizes the expected cost is an optimal action with prior information only. The minimum prior expected cost serves as a reference value to evaluate the value of an I-PR program.

The upper branch with inspection involves a Bayesian pre-posterior analysis considering all probable inspection results and computes the expected cost of optimal actions with inspection information. The difference between optimal expected costs with and without inspection is a measure of value of the I-PR program.

The value of I-PR program is derived from two aspects. First of all, an inspection will determine the actual magnitude of degradation, i.e., $X(t_1) = x_1$, which will change the distribution of future degradation, $X(t_2)$, that is required in the replacement decision. Second, the degradation observation also allows to update the prior distribution of the gamma process parameter. In essence, the inspection data in the context of a gamma process model would influence both temporal and parameter (or, epistemic) uncertainties. This paper presents an analytical formulation and computational approach to solve this decision problem.

The significance of using the gamma process as a model of degradation in the current VOI problem is discussed in Section 2.5. The present decision problem involves two types of uncertainties, viz., parameter and temporal, which are defined in clear terms in Section 2.2 and additional aspects of these uncertainties are discussed in Sections 2.2.1 and 2.4.3, respectively. Key assumptions of the probabilistic analysis and inspection policy are listed in Section 3.2.

1.4. Related literature

Traditionally, the optimal inspection and maintenance problem was examined from a probabilistic life-cycle cost perspective [1–3]. The stochastic gamma process has been widely used as mode of degradation in inspection and maintenance optimization problems. Such problems are formulated in terms of the minimization of expected cost with

respect to the inspection interval and the threshold for preventive replacement [4]. However, these formulations do not consider the effect of actual degradation data, obtained by inspection, on reducing uncertainties associated with the probabilistic degradation model itself. The traditional expected cost approach does not address a more fundamental question: is it worth implementing an I-PR program in the first place? In essence, the decision problem related to inspection/maintenance of a system affected by a gamma degradation process has not been fully analysed.

The estimation of expected value generated by of an I-PR program can be properly formulated using the concept of the value of information (VOI), which was largely originated from the work of Howard [5]. In the context of structural health monitoring, VOI theory has been used to quantify the value of various inspection methods [6–11]. The value generated by a structural health monitoring system was studied in [12–14]. The VOI concept has found many applications in engineering, e.g., sequential inspection planning of offshore structures [15], oil and gas pipelines [16], bridges [17–19], ships [20], and pavement [21].

To reduce computational efforts associated with a VOI analysis, many studies had adopted simple degradation models (largely categorized as a random variable degradation model in the sense of [3]), with a few exceptions such as [11,21,22], where a sophisticated partially observable Markov decision process was employed.

Besides the computational developments for VOI analysis, Straub and Faber (2004) investigated the system effects and the relation between inspection quantity and quality using structural reliability models [7,8]. Konakli et al. (2016) demonstrated how the value of information is influenced by the assumed probabilistic models, the mean trend of the degradation law, the measurement error, and stochastic dependence between the components of a system [23]. The Bayesian approach for modelling the parameter uncertainty in gamma process was discussed in [24]. The study presented different types of priors for parameters of the gamma process, and formulated a Markov Chain Monte Carlo algorithm for the posterior analysis. Using a stationary gamma process degradation model, Fauriat and Zio (2020) studied optimal planning of inspection and maintenance actions, but the VOI analysis was based on the minimum average cost rate calculation for age-based replacement strategy.

Flexible maintenance strategies based on sequential decision making are presented in [25,26]. A recent development is the consideration of decision maker's risk tolerance in the quantification of VOI through the use of utility function [27,28].

A review of literature suggests that VOI in most studies is gained due largely to a change in parameter uncertainty associated with the model parameters. However, when degradation is modelled as a stochastic process, the temporal uncertainty also becomes a confounding element in the decision analysis [3,29]. By temporal uncertainty, it is meant that in spite of knowing the current state of degradation, the future degradation state remains a probabilistic variable due to aleatory uncertainty associated with the stochastic process. Therefore, degradation data collected via an inspection program or a health monitoring system has potential to generate additional value by changing the temporal uncertainty associated with the degradation process in inspected components.

To conclude, the total VOI in a stochastic process model is a result of an intricate interaction between parameter and temporal uncertainties associated with the process. This aspect of the VOI problem has not yet been fully investigated in the literature. Thus, the main contribution of the work is an analytical formulation of the Bayesian VOI problem involving the gamma process of degradation. This model helps delineate the different value components associated with temporal and parameter uncertainties of the problem.

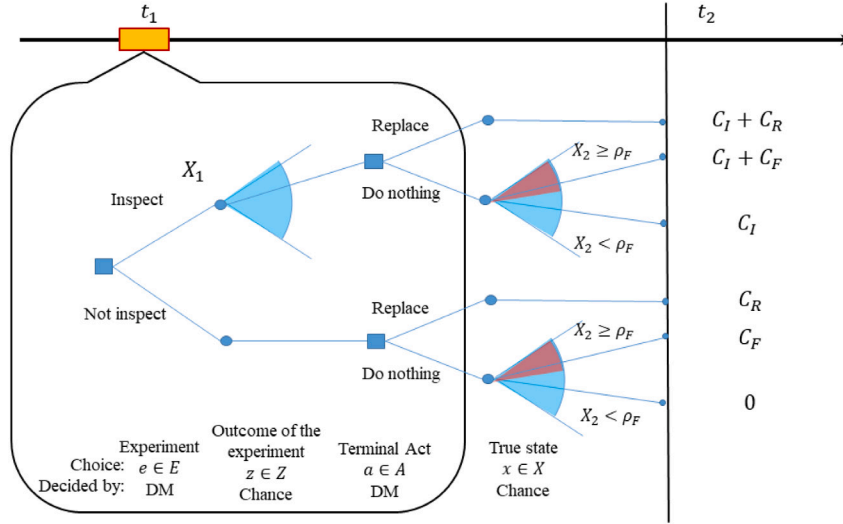


Fig. 1. The decision tree for inspection and preventive replacement of a component.

1.5. Organization of paper

The paper is organized as follows. First the Bayesian analysis of a stochastic gamma process is presented in Section 2. This formulation is applied to analyse a single-component inspection/replacement problem in Section 3, whereas Section 4 extends the analysis to a more practical, multiple-component inspection/replacement problem. Numerical examples with sensitivity analyses and an in-depth discussion of results are presented in Section 5. The effects of neglecting either the temporal or parameter uncertainty from the value analysis are discussed in Section 6. Conclusions of this study are presented in the last Section.

2. Gamma degradation process: Bayesian model

2.1. Notations

A random variable is denoted as usual by an upper case Roman letter and its specific realization by a lower case letter. Model parameters are denoted by Greek letters whenever possible. A conditional relationship is shown as $A|B$. In a mathematical function, a list of parameters is occasionally included after the primary variable separated by a semi-colon. This is intended to denote explicitly the parameters on which the function depends. Parameters related to the prior, posterior, and pre-posterior distributions are denoted by three primary superscripts, π , p , and $p\pi$, respectively.

Since this study involves two different time instances in both prior and posterior analyses, a subscript i ($i = 1, 2$, and Δ) is used to denote the time points, t_1 , t_2 , and the time difference $\Delta t = t_2 - t_1$, respectively. In addition, two composite superscripts, “ $p1$ ” and “ $p\pi$ ”, are used for posterior quantities derived from the single- and multiple-component sample, respectively.

Key notations of the formulation are summarized in Table 1 for ease of reference.

2.2. Basic model

Gamma process is a widely used degradation model in reliability and maintenance engineering. The present model includes two types of uncertainties that affect the formulation of VOI problem, viz., temporal and parameter uncertainties, which share characteristics of aleatory and epistemic uncertainties, respectively.

A key feature of gamma process is stationary, independent, and gamma distributed increments. As a result, the cumulative degradation,

Table 1

Key parameters and notations.

Parameters	Prior (π)	Posterior	
		Single-component	Multiple-component
Mean deterioration rate: $\mu \sim \text{IGa}(\alpha, \beta)$			
α	α	$\alpha^{p1} = \alpha + t_1/v^2$	$\alpha^{pn} = \alpha + nt_1/v^2$
β	β	$\beta^{p1} = \beta + x_1/v^2$	$\beta^{pn} = \beta + s_n/v^2$
Deterioration state: $X(t_i) \sim \xi_i F(2k_i, 2\alpha)$ ($i = 1, 2$)			
ξ_i	$\xi_i^\pi = \beta t_i/\alpha$	—	$\xi_i^{pn} = \beta^{pn} t_i/\alpha^{pn}$
k_i	$k_i = t_i/v^2$	—	$k_i = t_i/v^2$
Deterioration increment: $\Delta X(t_2) [= X(t_2) - x_1] \sim \xi_\Delta F(2k_\Delta, 2\alpha)$			
ξ_Δ	$\xi_\Delta^\pi = \beta \Delta t/\alpha$	$\xi_\Delta^{p1} = \beta^{p1} \Delta t/\alpha^{p1}$	$\xi_\Delta^{pn} = \beta^{pn} \Delta t/\alpha^{pn}$
k_Δ	$k_\Delta = \Delta t/v^2$	$k_\Delta = \Delta t/v^2$	$k_\Delta = \Delta t/v^2$
Sum deterioration: $S_m [= \sum_{j=1}^m X^{(j)}(t_1)] \sim \xi_m F(2k_m, 2\alpha)$ ($m = n - 1, n$)			
ξ_m	$\xi_m^\pi = m\beta t_1/\alpha$	—	—
k_m	$k_m = mt_1/v^2$	—	—
Probability of failure			
p_f	$p_f^\pi = \text{Eq. (14)}$	$p_f^{p1} = \text{Eq. (19)}$	$p_f^{pn} = \text{Eq. (31)}$

$X(t)$, over a time interval $(0, t]$, $t > 0$, follows the gamma distribution with the following probability density function (PDF):

$$g_{X(t)}(x) = \frac{x^{t/v^2-1}}{\Gamma(t/v^2)(\mu v^2)^{t/v^2}} \exp\left(-\frac{x/v^2}{\mu}\right), \quad (3)$$

which is parametrized in terms of the mean, μ , and coefficient of variation (COV), v , of the degradation increment in a unit time interval, i.e., $\Delta X = X(t+1) - X(t)$. With this parametrization, the scale and shape parameters of a stationary gamma process are given as μv^2 and t/v^2 , respectively [29]. A compact notation is frequently used to denote the gamma distribution as, $X(t) \sim \text{Ga}(\mu v^2, t/v^2)$.

The paper considers a planning horizon $[t_1, t_2]$, in which t_1 represents the time of decision. The mean, standard deviation and COV of cumulative degradation, $X(t_2)$, at the end of time horizon are given as

$$\mu_{X(t_2)} = \mu t_2, \quad \sigma_{X(t_2)} = (\mu v) \sqrt{t_2}, \quad v_{X(t_2)} = v/\sqrt{t_2}. \quad (4)$$

Since the current decision problem deals with components of surviving up to age t_1 , the distribution of degradation at t_2 takes a conditional form:

$$\Pr[X(t_2) \leq x | X(t_1) \leq \rho_F] = \frac{\Pr[X(t_2) \leq x, X(t_1) \leq \rho_F]}{\Pr[X(t_1) \leq \rho_F]}$$

$$= \frac{\int_0^{\rho_F} \text{Ga}(x-y; \mu v^2, \Delta t/v^2) g_{X(t_1)}(y) dy}{\text{Ga}(\rho_F; \mu v^2, t_1/v^2)}, \quad (5)$$

with $\Delta t = t_2 - t_1$. The mean and variation of this distribution cannot be computed explicitly.

Suppose an inspection at an intermediate time t_1 reveals, $X(t_1) = x_1$ ($0 < x_1 < \rho_F$), the conditional distribution of $X(t_2)$ can be derived using the independent increment property of the gamma process which allows to write, $X(t_2) = X(t_1) + \Delta X(\Delta t)$, where $\Delta X(\Delta t)$ is gamma distributed with the scale and shape parameters, μv^2 and $\Delta t/v^2$, respectively, and $\Delta t = t_2 - t_1$. Thus, the conditional cumulative distribution of $X(t_2)$ also follows a gamma distribution given as,

$$\begin{aligned} \Pr[X(t_2) \leq x_2 | X(t_1) = x_1] &= \Pr[X(t_1) + \Delta X \leq x_2 | X(t_1) = x_1] \\ &= \Pr[\Delta X \leq (x_2 - x_1)] \\ &= \text{Ga}(\Delta x; \mu v^2, \Delta t/v^2) \end{aligned} \quad (6)$$

where $\Delta x = x_2 - x_1$. The mean and variance of this distribution are given as $x_1 + \mu \Delta t$ and $(\mu v)^2 \Delta t$, respectively.

2.2.1. Discussion of temporal uncertainty

The gamma process model characterizes uncertainty associated with the magnitude of degradation, $X(t_1)$, at some point in time, t_1 . This uncertainty is of *aleatory* nature, as it is a result of random variations in the degradation process.

This paper uses a term, *temporal uncertainty*, to signify the dynamic nature of uncertainty associated with future evolution of degradation in a specific sample path of the process. Although temporal uncertainty is a consequence of aleatory uncertainty associated with the degradation process, there is a subtle distinction between them when an inspection is involved in the problem. The reason is that an inspection observation could change the temporal uncertainty.

In the present context, the degradation process in a given component is essentially a sample path of a gamma process. For example, degradation values at times, $t_1 < t_2 < \dots < t_n$, along a sample path of a gamma process are given by a vector of correlated random variables, $[X(t_1), X(t_2), \dots, X(t_n)]$. Therefore, given an observation, $X(t_1) = x_1$, the remaining variables, $X(t_2), \dots, X(t_n)$, would follow a conditional joint distribution which would be different from the original joint distribution.

In absence of an inspection, uncertainty associated with $X(t_2)$ is given by a gamma distribution with a variance, $(\mu v)^2 t_2$. However, an observation, $X(t_1) = x_1$, reduces the variance of $X(t_2)$ to $(\mu v)^2 (t_2 - t_1)$, as shown by Eq. (6). In addition, the mean and COV could also increase or decrease depending on values of x_1 and t_1 . This is referred to as a change in or updating of temporal uncertainty by inspection.

This change in temporal uncertainty by inspection is also applicable to a component of age t_1 for which distribution of $X(t_2)$ before inspection is given by Eq. (5), though there are no explicit formula for its moments.

In summary, although it is well known that aleatory uncertainty associated with a random experiment does not change by inspection, the temporal uncertainty refers to uncertainty at multiple points in time in future that is described by a joint distribution of vector of random variables. Therefore, observations about a subset of variables would change the distribution of remaining variables via conditioning. The degree of change would depend on the degree of correlation among random variables of the vector.

2.2.2. Parameter uncertainty

In the gamma process model used in this study, the parameter, μ , is treated as an uncertain parameter, whereas the COV (v) is taken as a known constant. The prior information about μ is specified by an inverse gamma distribution with PDF:

$$\pi(\mu) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{\mu}\right)^{\alpha+1} \exp\left(-\frac{\beta}{\mu}\right), \quad (7)$$

where α and β are hyper parameters of the prior, which can be determined from available information, or experts' knowledge. In shorthand, $\mu \sim \text{IGa}(\alpha, \beta)$. An inverse gamma distribution has the following mean (μ_π), standard deviation (σ_π), and COV (v_π):

$$\mu_\pi = \frac{\beta}{(\alpha-1)}, \quad \sigma_\pi = \frac{\beta}{(\alpha-1)\sqrt{\alpha-2}}, \quad \text{and} \quad v_\pi = \frac{1}{\sqrt{\alpha-2}}. \quad (8)$$

With a conjugate prior for μ , the predictive distribution of degradation, $X(t)$, can be derived as

$$\begin{aligned} \pi_X(x; t, \alpha, \beta) &= \int_0^\infty \text{Ga}(x; \mu v^2, t/v^2) \cdot \text{IGa}(\mu; \alpha, \beta) d\mu \\ &= \frac{1}{x B(t/v^2, \alpha)} \frac{\beta^\alpha (x/v^2)^{t/v^2}}{(\beta + x/v^2)^{\alpha+t/v^2}}, \end{aligned} \quad (9)$$

where $B(\alpha, t/v^2)$ is the standard Beta function. Note that the PDF of a regular F-distribution with degrees of freedom, $(2k, 2\alpha)$, has the following form,

$$f(y; 2k, 2\alpha) = \frac{1}{y B(k, \alpha)} \frac{(ky)^k \alpha^\alpha}{(ky + \alpha)^{(k+\alpha)}}. \quad (10)$$

With

$$\xi = \beta t / \alpha, \quad k = t / v^2, \quad (11)$$

it can be shown that $Y = X(t)/\xi$ follows the F-distribution, or $X(t)$ is a scaled F-distribution and ξ is the scale parameter. The shorthand notation $X(t) \sim \xi F(2k, 2\alpha)$ is used hereafter. The marginal cumulative distribution of $X(t)$ is thus given as

$$\Pr[X(t) \leq x] = F(x/\xi; 2k, 2\alpha) = \int_0^{x/\xi} f(y; 2k, 2\alpha) dy. \quad (12)$$

2.3. Probability of failure: No inspection data

A component failure occurs in an interval, $(0, t]$, if the cumulative degradation, $X(t)$, exceeds a critical threshold, ρ_F . Since $X(t)$ is a monotonic stochastic process, the probability of failure in an interval, $(0, t]$ can be evaluated using Eq. (12) as:

$$p_f = \Pr[X(t) > \rho_F] = 1 - F(\rho_F/\xi; 2k, 2\alpha). \quad (13)$$

Since this paper deals with a population of components that have survived up to age t_1 at the time of inspection, which implies that potential degradation in components is less than the failure limit, i.e., $X(t_1) < \rho_F$. In the calculation of probability of failure in a future interval, $(t_1, t_2]$, this information must be included via conditioning as follows:

$$\begin{aligned} p_f &= \Pr[X(t_2) > \rho_F | X(t_1) \leq \rho_F] \\ &= 1 - \Pr[X(t_2) \leq \rho_F | X(t_1) \leq \rho_F] \\ &= 1 - \Pr[X(t_2) \leq \rho_F, X(t_1) \leq \rho_F] / \Pr[X(t_1) \leq \rho_F] \\ &= 1 - \Pr[X(t_2) \leq \rho_F] / \Pr[X(t_1) \leq \rho_F]. \end{aligned}$$

The last equality holds true because $X(t_1) \leq \rho_F$ is a subset of $X(t_2) \leq \rho_F$. Substituting Eq. (13) into this relation leads to the following expression for the prior predictive probability of failure in the interval, $(t_1, t_2]$,

$$p_f^\pi = 1 - \frac{F(\rho_F/\xi_2^\pi; 2k_2, 2\alpha)}{F(\rho_F/\xi_1^\pi; 2k_1, 2\alpha)}, \quad (14)$$

in which, according to Eq. (11), $\xi_i^\pi = \beta t_i / \alpha$, $k_i = t_i / v^2$ for $i = 1, 2$. Thus, the survival of a component at time t_1 provides an extra piece of information: $X(t_1) \leq \rho_F$, which is duly accounted in Eq. (14).

Note that in case of an inspection this survival information need not be used in updating the prior distribution of μ . The reason is that an inspection will provide more precise information about the actual magnitude of degradation, i.e., $X(t_1) = x_1$, which would make the information, $X(t_1) \leq \rho_F$, superfluous for the posterior updating.

2.3.1. Summary and interpretation of model parameters

The degradation model associated with the present decision problem uses the following parameters to model parameter and temporal uncertainties discussed in the previous Section:

1. The gamma process model involves two parameters, viz., μ and ν , which are the mean and COV of annual degradation increment. The COV, ν , is an indicator of variability in the degradation increment. A case of $\nu = 1$ implies that the degradation increment follows an exponential distribution. The variance of annual degradation increment is given as $(\mu\nu)^2$, which affects the temporal uncertainty associated with the gamma process in a given time interval.
2. The mean parameter, μ , is treated uncertain to model the parameter uncertainty using an inverse gamma distribution with two additional parameters, α and β (see Eq. (7)), or the mean (μ_π) and COV (ν_π) of the prior, as shown by Eq. (8). The prior COV, ν_π , is an indicator of uncertainty associated with μ , whereas the prior mean, μ_π , indicates the rate of degradation (high or low). Parameter uncertainty, as reflected by ν_π , has a great deal of influence on the VOI decision problem, as shown by examples given in Section 5.

This set of three model parameters, ν, μ_π, ν_π , can be estimated by available data as well as degree of belief of the analyst.

2.4. Bayesian updating with inspection data

2.4.1. Single component inspection

An inspection at time t_1 provides information about the cumulative degradation as $X(t_1) = x_1$, which can be used to update the distribution of uncertain mean, μ . Since the inverse gamma distribution is a conjugate prior, the posterior of μ , $p(\mu|x_1)$, is also an inverse gamma, i.e. $(\mu|x_1) \sim \text{IGa}(\alpha^{p1}, \beta^{p1})$, with the following parameters:

$$\alpha^{p1} = \alpha + \frac{t_1}{\nu^2}, \quad \beta^{p1} = \beta + \frac{x_1}{\nu^2}. \quad (15)$$

Given an observation, $X(t_1) = x_1$, the predictive distribution of degradation $X(t_2)$, can be obtained as,

$$f(x_2|x_1) = \int_0^\infty f(x_2, \mu|x_1) d\mu = \int_0^\infty f(x_2|\mu, x_1) p(\mu|x_1) d\mu \quad (16)$$

where $f(x_2|\mu, x_1)$ follows a gamma density corresponding to the distribution given by Eq. (6), and $p(\mu|x_1)$ is the posterior inverse gamma distribution. Using these relations, the posterior predictive distribution of $X(t_2)$ is derived, similar to the derivation of Eq. (9), as

$$p(x_2|x_1) = \frac{1}{\Delta x B(\alpha + t_1/\nu^2, \Delta t/\nu^2)} \frac{(\beta + x_1/\nu^2)^{\alpha+t_1/\nu^2} (\Delta x/\nu^2)^{\Delta t/\nu^2}}{(\beta + x_2/\nu^2)^{\alpha+t_2/\nu^2}}. \quad (17)$$

This is the PDF of a scaled F distribution: $\xi_D^{p1} F(2k_D, 2\alpha^{p1})$, in which α^{p1} is given by Eq. (15), and the other two parameters are defined as follows:

$$\xi_D^{p1} = \beta^{p1} \Delta t / \alpha^{p1}, \quad k_D = \Delta t / \nu^2. \quad (18)$$

In other words, while the prior predictive distribution of $X(t_2)$ is a scaled F-distribution $X(t_2) \sim \xi_D^\pi F(2k_D, 2\alpha)$, the posterior predictive distribution of $\Delta X(t_2) = X(t_2) - x_1$ with a given x_1 is also a scaled F-distribution: $\Delta X(t_2) \sim \xi_D^{p1} F(2k_D, 2\alpha^{p1})$.

The posterior failure probability at time t_2 conditioned on $X(t_1) = x_1$ is thus given as

$$\begin{aligned} p_f^{p1} &= \Pr[X(t_2) > \rho_F | X(t_1) = x_1] \\ &= 1 - F_{2k_D, 2\alpha^{p1}}[(\rho_F - x_1)/\xi_D^{p1}]. \end{aligned} \quad (19)$$

2.4.2. Multiple-component inspection

Inspection of n components at time t_1 would provide a random sample of gamma distributed degradation values, which are denoted by a vector, $\mathbf{x} = (x_1^{(1)}, x_1^{(2)}, \dots, x_1^{(n)})$. Since a gamma distributed random sample has a sufficient statistics, $s_n = \sum_{j=1}^n x_1^{(j)}$, actual sample values are not needed to update the prior distribution.

Thus, the posterior of the mean of gamma process can be obtained similar to Eq. (15) as $(\mu|\mathbf{x}) = (\mu|s_n) \sim \text{IGa}(\mu; \alpha^{pn}, \beta^{pn})$, in which

$$\alpha^{pn} = \alpha + nt_1/\nu^2, \quad \beta^{pn} = \beta + s_n/\nu^2 \quad (20)$$

In summary, the concept of sufficient statistics, s_n , along with an inverse gamma conjugate prior permits a complete analytical formulation of the Bayesian analysis, and the subsequent pre-posterior problem.

2.4.3. Discussion of parameter uncertainty

In the Bayesian modelling, uncertainty arising from the lack of knowledge about a parameter is formally expressed by assigning it a probability distribution over a plausible set of values. In the present VOI problem, a parameter of gamma process, μ , is treated uncertain with an inverse gamma prior distribution, as given by Eq. (7).

Since uncertainty arising from lack of knowledge is also referred to as *epistemic uncertainty*, the parameter uncertainty is equivalent to epistemic uncertainty [30]. This paper uses the term *parameter uncertainty* as it is more specifically related to a model parameter, whereas “epistemic” uncertainty is a broad term that can include other types, e.g., model uncertainty. For a more detailed discussion of impact of parameter uncertainty on reliability, readers are referred to [31].

The parameter uncertainty is also updated by an inspection. As shown by Eq. (15), both posterior parameters of the inverse gamma prior distribution always increase given an observation, $X(t_1) = x_1$. Because of this the variance of the prior can increase or decrease depending on values of x_1 and t_1 , as shown by Eq. (8). However, the COV of prior, ν_π , will always decrease given an inspection observation.

In summary, establishing a clear distinction between aleatory and epistemic uncertainties is a challenging and widely debated issue in the literature. As discussed in [30], engineering reliability, risk and decision problems are generally solved within the confines of a model universe. Therefore, a characterization of problem uncertainties is only relevant within the confines of the defined model.

2.5. Role of degradation models in the VOI problem

The formulation of current decision problem depends on conceptual modelling of the degradation process, which can be classified into two broad categories: random variable (RV) and random process (stochastic) models, as discussed in [3] in the context of probabilistic modelling of system inspection and maintenance.

In the first RV model, degradation sample paths are described by a function of time with parameters that are modelled as time-invariant, random variables. An important aspect of this model is that a particular sample path can be made completely deterministic in time by an appropriate number of inspections. For example, consider a simple linear model, $X(t) = Rt$, with a random rate, R , as shown in Fig. 2a. In this case, a single, perfect inspection at time t_1 would reveal the actual magnitude of degradation as $X(t) = x_1$ using which the rate can be calculated as $r_1 = x_1/t_1$. Once the rate is known, all future values of degradation become fully deterministic, e.g., $X(t_2) = r_1 t_2 = x_2$. Therefore, a correct decision to “replace”, or “do nothing”, can be taken at time t_1 with certainty. Consequently, the VOI problem can be simply formulated as an expected value of perfect information (EVPI), which is somewhat elementary in nature and not pursued in this paper.

In case of a single sample path of gamma process, degradation values at various time points are described a vector of correlated random variables. Therefore, even a perfect inspection at time t_1 does not eliminate all the uncertainty associated with future values of degradation. For example, Fig. 2b shows a sample path of a stationary

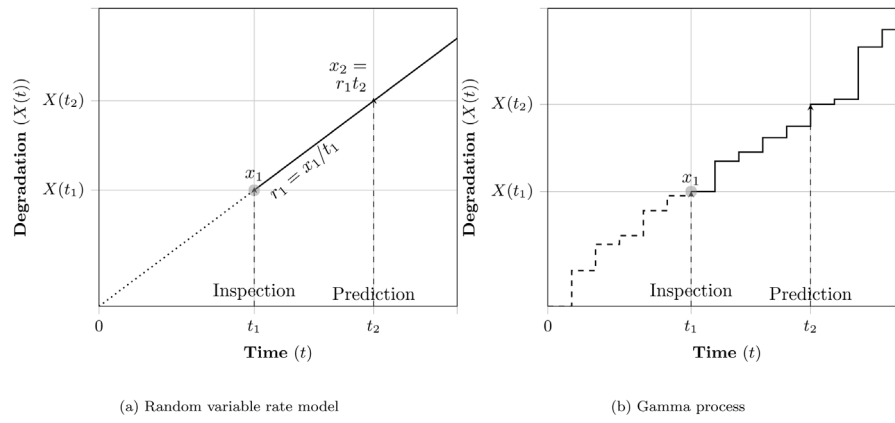


Fig. 2. Distinct nature of sample paths in two classes of degradation models.

gamma process, a jump process, which does not follow any parametric curve. Because of this, $X(t_2)$ remains uncertain in spite of an observation $X(t_1) = x_1$, and this uncertainty is described by a conditional distribution, (6).

In summary, a perfect inspection along a sample path of a gamma process does not reduce the VOI problem to a simplified form of EVPI, and the temporal uncertainty along a sample path continues to confound the decision problem.

3. The single-component problem

3.1. Inspection policy

This problem is concerned with a single component which has age t_1 at the time of inspection, and it is safely operating up to this time. Inspection cost is assumed to be C_I per component, and the cost of a preventive replacement as C_R . If the component is not replaced at inspection time t_1 , it may fail during the subsequent operation window, $(t_1, t_2]$. The replacement of a failed component during the operation is referred to as a “corrective” replacement. The cost of a corrective replacement, C_F , is often much higher than C_R .

3.2. Key assumptions

Key assumptions of the probabilistic analysis and inspection policy are as follows:

1. The present formulation ignores uncertainties involved with inspection data, such as measurement errors, as the principal aim is to investigate interactive effects of temporal and parameter uncertainties on the value of information in an analytical manner. Consideration of additional inspection uncertainties would further complicate the analysis.

Although measurement error is a common aspect of inspection tools, modern technology and instrumentation has considerably reduced its impact, especially in the inspection of wall thickness of metallic components. For example, an ultrasonic probe can measure the pipe wall thickness with high accuracy with negligible random noise, or measurement error [32]. This finding was based on an analysis of a large database of nuclear piping thickness measurements. As the present formulation will be applied to an example of corrosion management in metallic pipes, ignoring measurement error is justified. However, in measuring properties of more heterogeneous material like concrete, the measurement error could be significant and it should be duly included in the analysis.

2. After an installation of a component at time zero, only a single replacement is permitted, whether it is done preventively at an inspection, or correctively upon a failure. In other words, possible renewals of components within the planning horizon $[t_1, t_2]$ are not considered in the analysis.

This assumption avoids the use of a more complex, stochastic renewal model in the analysis. This assumption can be justified on the grounds that the expected lifetime of a component is much longer than the remaining operational period, which implies that the probability of failure of a replaced component is fairly small in the remaining service life.

3. The failure is assumed to take place in a self-announced mode. It means that any failure in an operation window, $[t_1, t_2]$, would be detected and subjected to a corrective replacement without any delay. An example of such a failure mode is a leak in a pipe section.

3.3. Prior analysis

First of all, in the absence of an inspection program, the plant operator at time t_1 has only two options: either to replace the component preventively, or do nothing, as shown by the bottom branch of the decision tree in Fig. 1. The cost of preventive replacement is C_R with certainty, whereas the “do nothing” option may cost C_F only if the component fails between t_1 and t_2 .

According to statistical decision theory, the optimum action is the one that minimizes the expected cost over the time interval, $[t_1, t_2]$, which is given as

$$C_1^\pi = \min \left(C_R, p_f^\pi C_F \right), \quad (21)$$

in which p_f^π given by Eq. (14) is the probability of failure evaluated under the prior information only.

3.4. Posterior analysis

When an inspection provides a data point, $X(t_1) = x_1 < \rho_F$, it can be used to update the prior of μ , and compute the posterior probability of failure, denoted as $p_f^{p1}(x_1)$, as given by Eq. (19).

The optimal replacement decision rule is thus similar to that without inspection, i.e., if $p_f^{p1}(x_1) > C_R/C_F$, then replace the component preventively; otherwise no action is required.

Alternatively, this decision rule can be expressed in terms of a directly observable x_1 . Substituting Eq. (19) into the break-even condition of failure probability $p_f^{p1}(x_1) = C_R/C_F$ and simple rearrangement yields

$$F_{2k_A, 2\alpha^{p1}} \left[\frac{\rho_F - x_1}{\xi_A^{p1}} \right] = 1 - \frac{C_R}{C_F} \quad (22)$$

Solving this equation for x_1 yields the preventive replacement (PR) threshold, x_{1r} . Based on this, the following decision rule is proposed:

$$a^{p1}(t_1) = \begin{cases} a_1 : \text{Replace,} & \text{if } x_1 \geq x_{1r}; \\ a_2 : \text{Do not replace,} & \text{if } x_1 < x_{1r} \end{cases} \quad (23)$$

Comparing to the prior analysis in the above subsection, the PR threshold suggests a true condition-based replacement policy.

Since the inspection cost has become a sunk cost (i.e., a cost spent that will not affect the present decision), the posterior expected cost is thus expressed as follows:

$$\begin{aligned} C_1^p(x_1) &= C_I + \min\{C_R, p_f^{p1}(x_1)C_F\} \\ &= \begin{cases} C_I + C_R, & \text{if } x_1 \geq x_{1r} \\ C_I + p_f^{p1}(x_1)C_F, & \text{otherwise.} \end{cases} \end{aligned} \quad (24)$$

3.5. Pre-posterior analysis

The pre-posterior analysis evaluates the minimum expected cost, $C_1^p(x_1)$, given by Eq. (24) for all possible values of degradation, $X(t_1) \leq \rho_F$, that could be observed in an inspection. Note that if the inspection finds a failed component, i.e., $X(t_1) = x_1 > \rho_F$, it must be replaced without the need for any decision analysis. Therefore, the prior predictive distribution of $X(t_1)$ is a truncated version of $\pi(x_1)$ given by Eq. (9). Denote this by $\pi'(x_1)$. Then $\pi'(x_1) = \pi(x_1)/F_{2k_1, 2\alpha}(x_{1r}/\xi_1^\pi)$ for $0 < x_1 < \rho_F$, and 0 otherwise.

Thus, the pre-posterior expected cost $C_1^{p\pi}$, i.e., the prior expected value of the posterior expected cost $C_1^p(x_1)$, is computed with respect to $X(t_1)$ as follows:

$$\begin{aligned} C_1^{p\pi} &= \int_0^{\rho_F} C_1^p(x_1) \pi'(x_1) dx_1 \\ &= C_I + p_R^{p1\pi} C_R + p_f^{p1\pi} C_F \end{aligned} \quad (25)$$

The two probabilities in the equation are evaluated as follows:

$$p_R^{p1\pi} = \int_{x_{1r}}^{\rho_F} \pi'(x_1) dx_1 = 1 - \frac{F_{2k_1, 2\alpha}(x_{1r}/\xi_1^\pi)}{F_{2k_1, 2\alpha}(\rho_F/\xi_1^\pi)} \quad (26)$$

$$p_f^{p1\pi} = \int_0^{x_{1r}} p_f^{p1}(x_1) \pi'(x_1) dx_1 \quad (27)$$

Here, $p_R^{p1\pi}$ represents the pre-posterior estimate of the replacement probability of the component (i.e., the probability that the replacement threshold is exceeded, which leads to a replacement), whereas $p_f^{p1\pi}$ stands for the pre-posterior average failure probability for a component that is inspected but not required to be replaced as per the break-even condition. The probability $p_f^{p1\pi}$ was evaluated by simulating samples from the truncated, scaled F-distribution of $\pi'(x_1)$.

3.6. Value of the I-PR program

By comparing the pre-posterior expected cost given by Eq. (25) with the prior expected cost expressed in Eq. (21), the expected value of an inspection and preventive replacement (I-PR) program can be evaluated. The value generated by an I-PR program, V_1 , is essentially equivalent to the Bayesian measure of the expected net gain of sampling (ENGs) given as

$$V_1 = C_1^\pi - C_1^{p\pi} = C_1^\pi - (C_I + p_R^{p1\pi} C_R + p_f^{p1\pi} C_F) \quad (28)$$

A negative value of ENGs means that the value generated by an I-PR program is insufficient to compensate for the cost of inspection that is required to get the information. Therefore, it is not worth undertaking such a program.

It is worth emphasizing that changes in temporal and parameter uncertainties in the light of inspection information do not generate a value per se. Rather, an information generates a value only when it leads to a change in the decision as compared to the prior optimal decision without inspection information. Such a change in the decision

occurs when there is an appreciable change in the posterior distribution of future degradation including a shift in its mean value as well as an increase or decrease in the variance. No value is generated in spite of changes in uncertainties, if new information ends up confirming the prior optimal decision.

4. Multiple-component problem

In this problem, the objective is to evaluate the value of implementing an I-PR program for a group of n statistically identical components. The structure of the decision problem is identical to that described in Fig. 1, and the formulation uses all the assumptions given in Section 3.2. An additional assumption is made that each component is affected by a gamma degradation process that is conditionally identical and independent from other components.

In principle, the prior and posterior analyses are similar to the single component problem, except that the pre-posterior analysis requires a more careful updating of the future failure probability of a component in the sample, and the break-even condition has to be formulated considering the entire sample of data. These aspects are clarified in Section 4.3.

4.1. Prior analysis

Since given a value of μ , degradation process in every component in the group is statistically independent and identical, the prior analysis for each of them is identical to the single component prior analysis discussed in Section 3.3. Thus, the prior optimal decision for every component is the same as that given by Eq. (21). The total prior expected cost for n components is therefore given as

$$C_n^\pi = n C_1^\pi \quad (29)$$

4.2. Posterior analysis

The posterior analysis of the multiple-component problem is analytically similar to that of the single-component problem, except that there are more observations for the Bayesian updating of the inverse gamma prior.

Given a sample of degradation data, $\mathbf{x} = (x_1^{(1)}, \dots, x_1^{(j)}, \dots, x_1^{(n)})$, the posterior of the mean of gamma process is easily obtained using the sufficient statistic of the sample, s_n , as given by Eq. (20). Correspondingly, the posterior predictive distribution of the future deterioration increment follows an updated scaled F-distribution, i.e., $\Delta X(t_2) \sim \xi_\Delta^{pn} F(2k_\Delta, 2\alpha^{pn})$, whose PDF is given as follows:

$$p(x_2^{(j)} | x_1^{(j)}, s_n) = \frac{1}{\Delta x^{(j)} B\left(\alpha + \frac{nt_1}{v^2}, \frac{\Delta t}{v^2}\right)} \frac{\left(\beta + \frac{s_n}{v^2}\right)^{\alpha + \frac{nt_1}{v^2}} \left(\frac{\Delta x^{(j)}}{v^2}\right)^{\frac{\Delta t}{v^2}}}{\left(\beta + \frac{s_n + \Delta x^{(j)}}{v^2}\right)^{\alpha + \frac{t_2 + (n-1)t_1}{v^2}}} \quad (30)$$

where $\Delta x^{(j)} = x_2^{(j)} - x_1^{(j)}$. This is the PDF of the scaled F-distribution $\xi_\Delta^{pn} F(2k_\Delta, 2\alpha^{pn})$, in which $\xi_\Delta^{pn} = \beta^{pn} \Delta t / \alpha^{pn}$.

This distribution is then used to predict the posterior probability of failure for the j th inspected component, $p_f^{pn}(x_1^{(j)}, s_n)$, as given below:

$$p_f^{pn}(x_1^{(j)}, s_n) = 1 - F_{2k_\Delta, 2\alpha^{pn}}\left[(\rho_F - x_1^{(j)})/\xi_\Delta^{pn}\right] \quad (31)$$

Following the same decision criterion as before, an inspected component will be replaced if $p_f^{pn} \geq C_R/C_F$, and not replaced otherwise.

Unlike the single-component problem for which the decision rule is governed by the univariate x_1 , the break-even condition for component j in the multi-component problem is a function of both $x_1^{(j)}$, the observed degradation of the j th component and s_n , the sum of observed degradation in all inspected components. Therefore, the break-even condition is rather a demarcation curve, which divides the $x_1^{(j)}-s_n$ space into a “replacement” zone and a “non-replacement” zone. This curve is influenced by many factors such as C_R , C_F , t_1 , t_2 , v , n , α and β .

Denoting the “replacement” zone by Ω_R and “non-replacement” zone by Ω_R^c , the decision rule for all the inspected components is given as follows:

$$a^{pn}(x_1^{(j)}, s_n) = \begin{cases} a_1 : \text{Replace,} & \text{if } (x_1^{(j)}, s_n) \in \Omega_R; \\ a_2 : \text{Do not replace,} & \text{otherwise.} \end{cases} \quad (32)$$

Correspondingly, the posterior expected cost of a component j is expressed as follows:

$$\begin{aligned} C_n^p(x_1^{(j)}, s_n) &= C_I + \min\{C_R, p_f^{pn}(x_1^{(j)}, s_n)C_F\} \\ &= \begin{cases} C_I + C_R, & \text{if } (x_1^{(j)}, s_n) \in \Omega_R; \\ C_I + p_f^{pn}(x_1^{(j)}, s_n)C_F, & \text{otherwise.} \end{cases} \end{aligned} \quad (33)$$

The total expected cost is computed as

$$C_n^p(\mathbf{x}) = nC_I + \sum_{j=1}^n \min\{C_R, p_f^{pn}(x_1^{(j)}, s_n)C_F\} \quad (34)$$

4.3. Pre-posterior analyses

The total pre-posterior expected cost is computed by taking an expectation over a random vector of potential degradation values, $\mathbf{x} = (x_1^{(1)}, \dots, x_1^{(n)})$, as

$$\begin{aligned} C_n^{p\pi} &= \int C_n^p(\mathbf{x})\pi(\mathbf{x})d\mathbf{x} \\ &= nC_I + \int \pi(\mathbf{x}) \sum_{j=1}^n \min\{C_R, p_f^{pn}(x_1^{(j)}, s_n)C_F\} d\mathbf{x} \end{aligned} \quad (35)$$

where $\pi(\mathbf{x})$ represents the joint prior PDF of the degradation of all components at the time of inspection.

Conceptually, substituting Eq. (34) into (35) would yield the total pre-posterior expected cost, but this involves a high-dimensional integration. Although simulation methods can be used to evaluate the integration, a simplified method is proposed, which is not only more efficient, but also reveals many insightful features of the VOI analysis for a stochastic process model. These aspects are discussed next in more details.

4.3.1. Intricacies of a stochastic VOI problem

A sample of degradation data, $\mathbf{x}$, provides the following two pieces of information for the replacement decision:

1. a sufficient statistic, s_n , that is used to reduce the epistemic, parameter uncertainty through Bayesian updating, as given by Eq. (20), and
2. the individual degradation, $x_1^{(j)}$, measured in a j th component that is used to reduce the aleatory uncertainty in the prediction of the failure probability of this component over a future time interval, $(t_1, t_2]$. As shown in Eq. (35), this calculation must be repeated for all $j = 1, \dots, n$.

Because of this inter-dependence, main challenges of a stochastic pre-posterior analysis are:

- How to consider all possible values of inspection data described by, $(x_1^{(j)}, s_n)$, $j = 1, n$, in the computation of expected cost of optimal actions?
- How to consider dependence in a tri-variate vector of random variables, $(X_1^{(j)}, S_n, \mu)$ that describes the pre-posterior analysis?
- How to determine an optimal action being replace or do not replace a component in the population of n components, given a realization of $(x_1^{(j)}, s_n)$?

4.3.2. Proposed solution

The key point to addressing these challenges lies in the use of the partial sum, $S_{n-1}^{(-j)} = \sum_{i=1, i \neq j}^n X_1^{(i)}$. The reason is that $X_1^{(j)}$ and S_n are dependent, whereas given μ , $X_1^{(j)}$ and $S_{n-1}^{(-j)}$ are conditionally independent. More importantly, the joint prior distribution of $X_1^{(j)}$ and $S_{n-1}^{(-j)} = \sum_{i \neq j} X_1^{(i)}$ is given below:

$$\pi(x_1, s_{n-1}; t_1, \alpha, \beta) = \frac{\Gamma(\alpha + \frac{nt_1}{v^2})}{\Gamma(\alpha)\Gamma(\frac{t_1}{v^2})\Gamma(\frac{(n-1)t_1}{v^2})} \frac{\beta^\alpha \left(\frac{x_1}{v^2}\right)^{\frac{t_1}{v^2}} \left(\frac{s_{n-1}}{v^2}\right)^{\frac{(n-1)t_1}{v^2}}}{x_1 s_{n-1} \left(\beta + \frac{s_n}{v^2}\right)^{\alpha + \frac{nt_1}{v^2}}}, \quad (36)$$

which is the joint PDF of a scaled bivariate F-distribution. The derivation is similar to that of the prior predictive distribution for $X(t)$ given in Eq. (9). The only difference is the univariate distribution $\text{Ga}(x; \mu v^2, t/v^2)$ there being replaced by the bivariate distribution for X_1 and S_{n-1} , which is the product of $\text{Ga}(x_1; \mu v^2, t_1/v^2)$ and $\text{Ga}(s_{n-1}; \mu v^2, (n-1)t_1/v^2)$. Integrating out of the prior distribution of μ yields this scaled bivariate F-distribution.

As a result, the multi-dimensional integration of Eq. (35) can be simplified as the following:

$$\begin{aligned} &\int \pi(\mathbf{x}) \sum_{j=1}^n \min\{C_R, p_f^{pn}(x_1^{(j)}, s_n)C_F\} d\mathbf{x} \\ &= \sum_{j=1}^n \int \pi'(x_1^{(j)}, s_{n-1}^{(-j)}) \min\{C_R, p_f^{pn}(x_1^{(j)}, s_{n-1}^{(-j)})C_F\} dx_1^{(j)} ds_{n-1}^{(-j)} \\ &= n \int \pi'(x_1, s_{n-1}) \min\{C_R, p_f^{pn}(x_1, s_{n-1})C_F\} dx_1 ds_{n-1} \\ &= n \left(p_R^{pn} C_R + p_f^{pn} C_F \right), \end{aligned} \quad (37)$$

where, $\pi'(x_1, s_{n-1})$, similar to the univariate counterpart in Section 3.5, is the truncated version of $\pi(x_1, s_{n-1})$ given by Eq. (36), truncated at $X_1 < \rho_F$ and $S_{n-1} < (n-1)\rho_F$; and the other two probabilities are given as below:

$$p_R^{pn} = \int_{\Omega_R} \pi'(x_1, s_{n-1}) dx_1 ds_{n-1} \quad (38)$$

$$p_f^{pn} = \int_{\Omega_R^c} p_f^{pn}(x_1, s_{n-1}) \pi'(x_1, s_{n-1}) dx_1 ds_{n-1} \quad (39)$$

The first equality in Eq. (37) holds because (i) integral is a linear operator, and (ii) the probability p_f^{pn} as a function of $x_1^{(j)}$ and s_n can be recast as a function of $x_1^{(j)}$ and $s_{n-1}^{(-j)}$. The second equality makes use of the homogeneity nature of the population. Thus, the total pre-posterior expected cost are calculated as

$$C_n^{p\pi} = n \left(C_I + p_R^{pn} C_R + p_f^{pn} C_F \right) \quad (40)$$

4.3.3. Computational details

Note that the two probabilities in Eq. (40) requires evaluation of two bivariate integrals for which the Monte Carlo simulation method is used here. The computation can be considerably simplified by the fact that $X(t_1)$ and S_{n-1} are conditional independent given a value of μ . Thus, the joint distribution of the trivariate vector, (X_1, S_{n-1}, μ) , can be written as

$$f(x_1, s_{n-1}, \mu) = f(x_1, s_{n-1} | \mu) f(\mu) = f(x_1 | \mu) f(s_{n-1} | \mu) f(\mu) \quad (41)$$

First, a sample of μ is generated from $\text{IGa}(\alpha, \beta)$. Then, random samples of x_1 and s_{n-1} are simulated, respectively, from $\text{Ga}(t_1/v^2, \mu v^2)$ and $\text{Ga}((n-1)t_1/v^2, \mu v^2)$. Conditions of $x_1 < \rho_F$ and $s_{n-1} < \rho_F$ are employed to ensure surviving components at time t_1 . Finally, in the simulated vector, (x_1, s_{n-1}) , the relative frequency of those samples that fall inside the “replacement” zone, Ω_R , provides an estimate of p_R^{pn} . Similarly, the failure probability, p_f^{pn} , is estimated by the average of $p_f^{pn}(x_1, s_{n-1})$ samples found in the “non-replacement” zone, Ω_R^c .

4.4. Value of the I-PR program for a component population

The value of an I-PR program, V_n , in the multi-component problem is evaluated as

$$V_n = C_n^\pi - C_n^{p\pi} = n \times [C_1^\pi - (C_I + p_R^{p\pi} C_R + p_f^{p\pi} C_F)] \quad (42)$$

The average value of I-PR program per component, V_{avg} , can be computed as the total value V_n divided by the number of components n . Thus,

$$V_{avg} = C_1^\pi - (C_I + p_R^{p\pi} C_R + p_f^{p\pi} C_F) \quad (43)$$

The synergistic effect of inspecting n components can be evaluated by comparing the average ENGS with the ENGS in the single-component problem given by Eq. (28).

5. Numerical examples

This section presents a numerical example that is inspired by corrosion management in the primary cooling piping system of a nuclear plant [33]. The wall thinning by flow accelerated corrosion (FAC) is the main degradation mechanism that adversely affects the integrity of the first pipe bend section. As the plant is approaching its end of life in about 10 years of time, the plant operator is interested in finding out the value of implementing an I-PR program for a group of 50 critical pipe bends.

The following numerical parameters values are taken as the reference case: $n = 50$, cost of corrective maintenance, $C_F = \$100$, cost of preventive replacement, $C_R = \$20$, cost of inspection, $C_I = \$1$, inspection time, $t_1 = 20$ year, end of planing horizon, $t_2 = 30$ year, COV of the gamma process, $\nu = 1/3$, and degradation failure threshold, $\rho_F = 3.0$ mm.

The cost information provided here is only meaningful in a relative term. Based on inspection data from other reactors, the mean deterioration rate is estimated to be 0.08887 mm/year. The COV of the inverse gamma prior is taken as equal to the COV of the gamma process, i.e., $\nu_\pi = \nu = 1/3$. Based on this information the prior parameters are calculated as $\alpha = 11$ and $\beta = 0.8887$.

5.1. Prior analysis

The prior probability of failure for the reference case is computed as $p_f^\pi = 0.251$, which is greater than the cost ratio $C_R/C_F = 0.20$. According to the break-even condition, the preventive replacement of the pipe bend is an optimal (prior) decision in absence of any other information. This is applicable to both the single- and multiple-component problems. Therefore, $C_1^\pi = C_R = \$20$ for the reference case.

5.2. Single-component problem

This Section presents results of the pre-posterior analysis of the single component inspection problem. The break-even condition is computed in terms of the degradation magnitude as, $x_{1r} = 1.935$ mm, using Eq. (22). This replacement threshold is 64.5% of the failure threshold, $\rho_F = 3$ mm.

The probability of replacement in the pre-posterior analysis is calculated using Eq. (26) as $p_R^{p1\pi} = 0.288$. This means that considering all possible inspection outcomes (i.e., degradation values), there is 28.8% probability that a pipe bend would be replaced. Recall that the prior analysis in the previous subsection showed that a preventive replacement is a more cost efficient decision, whereas the pre-posterior analysis shows that the information gained by an inspection would recommend a replacement with a probability that is less than one third.

This finding is further substantiated by the pre-posterior failure probability, $p_f^{p1\pi} = 0.0092$ or 0.92%, computed using Eq. (27). Since

$p_f^{p1\pi}$ is much smaller than its prior estimate of $p_f^\pi = 25.1\%$, the value of potential information to be gained by an inspection is quite significant.

The pre-posterior analysis leads to the total expected cost of optimal actions as, $C_1^{p\pi} = C_I + p_R^{p1\pi} C_R + p_f^{p1\pi} C_F = 1 + 0.288 \times 20 + 0.0092 \times 100 = \7.68 . In contrast, the prior estimate of the minimum expected cost is $C_1^\pi = \min\{C_R, p_f^\pi C_F\} = \20 . Since $C_1^{p\pi} < C_1^\pi$, the implementation of an I-PR program would be a cost-optimal decision, as it would generate a value of $V_1 = C_1^\pi - C_1^{p\pi} = \12.3 .

The COV of inverse gamma prior, ν_π , is a surrogate for the quality of prior information available to the plant operator. Therefore, the sensitivity of V_1 to ν_π is presented in Fig. 3. Note that the COV values on the X -axis are plotted on a log scale.

Variations of the prior and preposterior estimates of the minimum expected cost are displayed in Fig. 3(a), in which the top red curve is the prior minimum expected cost without inspection $C_1^\pi = \min\{C_R, p_f^\pi C_F\}$. This curve is basically the prior mean failure cost curve $p_f^\pi C_F$ truncated at the value of the preventive replacement cost, $C_R = \$20$. The pre-posterior expected cost, $C_1^{p\pi}$, is shown as the thick black curve in the middle of the plot, and its three cost components are shown underneath. Among them, the replacement cost $p_R^{p1\pi} C_R$ contributes the most to $C_1^{p\pi}$, particularly when ν_π is greater than 0.1. When $\nu_\pi < 0.05$, the expected failure cost $p_f^{p1\pi} C_F$ takes a major share of the total cost, indicating that the inspection data has little effect on reducing the failure probability.

The trend of V_1 in Fig. 3(b) clearly shows that a positive net value can be achieved from an I-PR program as long as ν_π is not too small, i.e., $\nu_\pi > 0.03$. In addition, this figure also reveals that a relatively large value is gained for a reasonable range of values of prior COV, i.e., $0.1 < \nu_\pi < 1$. For a large prior COV, i.e., $\nu_\pi > 1$, both C_1^π and $C_1^{p\pi}$ decrease, and as a result, the ENGS also decreases. This non-monotonic phenomenon is attributed to the tail behaviour of the scaled F-distribution, as given in Eq. (14).

5.3. Multiple-component problem

The replacement decision rule for the multiple-component problem is defined by a two-dimensional break-even curve in x_1 and s_{n-1} , as shown in Fig. 4, for the reference case of $\nu_\pi = 1/3$ and $C_R/C_F = 0.2$. The break-even curve, almost like a straight line, divides the plane into replacement (Ω_R) and non-replacement (Ω_R^c) zones. In this plot, contours of the joint prior distribution of $X(t_1)$ and $S_{n-1}(t_1)$, which is a scaled bivariate F-distribution given in Eq. (36), are also superimposed. The probability density contours indicate that the bivariate prior is a highly correlated and skewed distribution. From this plot, the volume of the bivariate distribution in the replacement zone (Ω_R) is equivalent to the pre-posterior probability of replacement, $p_R^{p\pi}$, as given by Eq. (38).

For the reference case, the pre-posterior probabilities of preventive replacement and failure were computed as $p_R^{p\pi} = 0.282$ and $p_f^{p\pi} = 0.0066$. Based on these values, the average value (V_{avg}) per component of the population was computed as $\$12.7$, which is larger than that for a single component, i.e., $V_1 = \$12.3$. However, for a group of 50 components, the total synergistic effect is $50 \times (12.7 - 12.3) = \20 , which is not a negligible amount. This point is further discussed in the next Section.

5.3.1. The group effect

The group effect refers to an additional gain in the value of information achieved by inspection of multiple components in comparison to inspection of a single component. The value of information synergy in the Bayesian updating can be quantified by $\Delta V = V_{avg} - V_1$. Substituting the expressions for V_{avg} and V_1 given by Eqs. (28) and (43), respectively, leads to

$$\Delta V = (p_R^{p1\pi} - p_R^{p\pi}) C_R + (p_f^{p1\pi} - p_f^{p\pi}) C_F. \quad (44)$$

Fig. 5 shows the variation of ΔV against ν_π along with a breakdown of the additional gain into its constituent components. The value gained

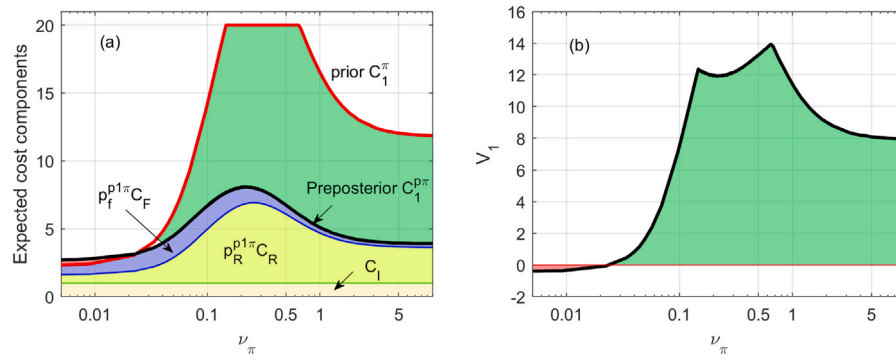


Fig. 3. The trends of (a) various constituents of the expected cost and (b) the value of I-PR program (V_1) along with the prior COV ν_π .

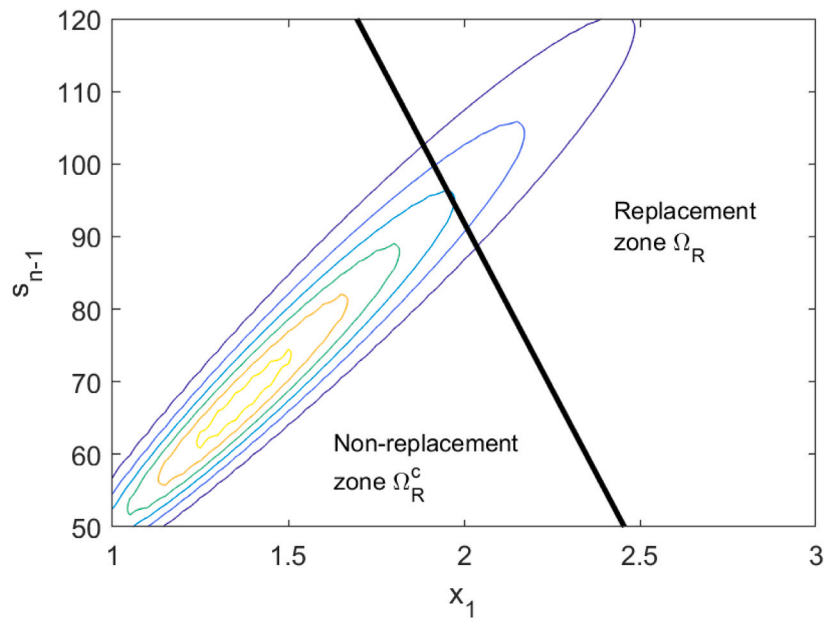


Fig. 4. The 2D decision region divided by the break-even curve.

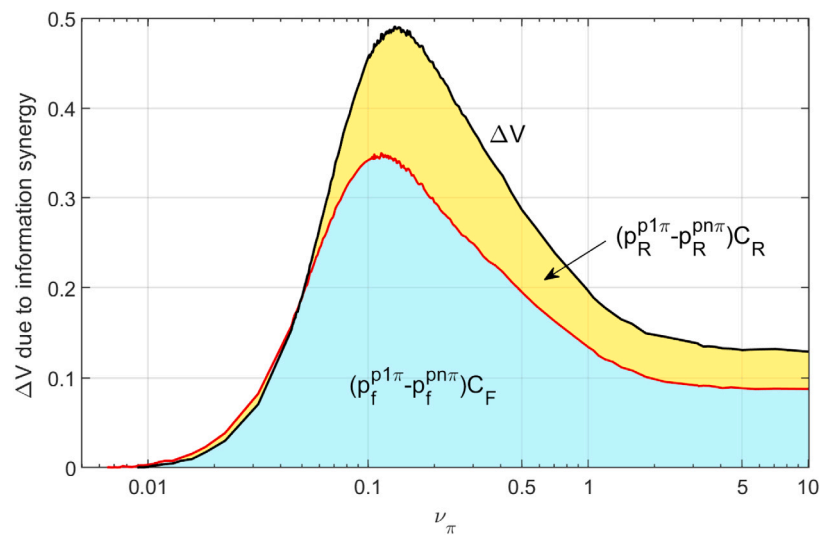


Fig. 5. The trend of ΔV with ν_π .

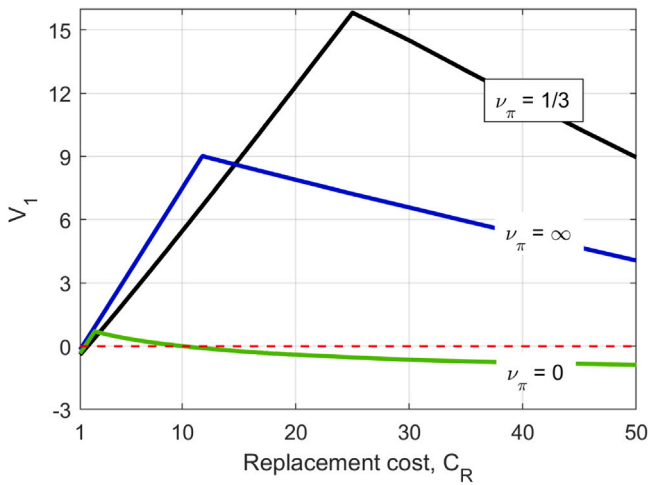


Fig. 6. Sensitivity of V_1 to C_R and v_π (with $C_I = 1$, $C_F = 100$).

by the group effect is positive for the entire range of v_π , though it starts to shrink to zero when $v_\pi < 0.01$. In addition, the group effect is quite pronounced in the middle range of variation, i.e., $0.1 < v_\pi < 0.5$.

However, a breakdown of the additional gain is quite informative. While the aggregated data collected from other components always help to reduce the failure cost, the additional information provided may slightly increase the mean number of replacements when v_π is less than 0.05. For the whole range of v_π , the additional value gained from information synergy is dominated by that resulting from a reduction in the failure probability.

5.4. Discussion

5.4.1. Cost ratio effect

The I-PR program involves three cost parameters: C_I , C_R and C_F . The effect of C_I on V_1 and V_{avg} is obvious: a one-unit increase in C_I will cause a one-unit reduction in both V_1 and V_{avg} . Therefore, this part of discussion focuses on the impact of C_R and C_F in terms of the cost ratio, C_R/C_F , because this ratio directly affects the decision rule in both prior and posterior analysis. Since the trends of V_1 and V_{avg} are qualitatively similar, as shown in Fig. 5, the following discussion uses only V_1 for illustration.

Fig. 6 shows V_1 against C_R for three v_π values. In all the three cases, V_1 increases almost linearly with an increasing C_R and reaches its peak when $C_R = p_f^* C_F$. For example, in the reference case of $v_\pi = 1/3$ and $C_R = p_f^* C_F = 25.1$, the value V_1 reaches the maximum of 15.9. The reason for this result is that when $C_R/C_F = p_f^*$, there is maximum uncertainty with the given prior information about what is an optimal action. Therefore, any information about the current degradation state will significantly improve the efficiency of a preventive replacement decision. For fairly low and significantly high values of C_R , there is no need for an inspection to support the decision making, because the preventive replacement would be a better decision in the former case, and “do nothing” in the latter.

Fig. 3 has already shown a non-monotonic relationship of V_1 with v_π for a fixed C_R value. Fig. 6 confirms this relationship over a wide range of C_R as well. More interestingly, when v_π goes to 0, at which there is no parameter uncertainty, the value V_1 reaches the smallest level among all v_π cases. In this case, the V_1 is gained purely from the elimination of the temporal uncertainty at time t_1 . Because of this, the value of inspection is limited and the maximum value is only about 0.75 at $C_R = \$2.3$. If C_I is greater than \$1.75, the whole V_1 curve will fall below zero and the inspection cannot be economically justified regardless of the C_R value. In other words, the utility of inspection is quite limited in absence of the parameter uncertainty from the gamma process model.

5.4.2. Impact of variability in the gamma process

A gamma process degradation model is characterized by its mean μ and COV v of the unit degradation increment or degradation rate. In this study, only uncertainty in the mean degradation rate is modelled by an inverse gamma distribution, and the degree of parameter uncertainty is measured by v_π as explained above. By contrast, the degree of temporal uncertainty is measured by the COV v of the degradation rate, which will also affect the results of the pre-posterior analysis, as seen from Eqs. (15) and (20).

Fig. 7 shows the effects of v on the variation of (a) V_1 and (b) ΔV . Clearly, the value of the I-PR program for the single-component problem decreases as the COV of the gamma process increases from 0.1 to 3. This is because in the single-component problem, a data with greater inherent temporal uncertainty, as indicated by a large value of v , makes a smaller contribution to the reduction in the parameter uncertainty (measured by v_π or α). On the other hand, the group effect measured by the ΔV shows an opposite trend with v , i.e., it increases with increasing values of v . This is because the group effect is generated by the reduction of the parameter uncertainty. As the degree of variability in gamma process increases, the same deterioration data provides relatively higher amount of information for the model parameter, resulting in a greater reduction in the parameter uncertainty.

6. Effect of simplifying the VOI analysis

The VOI analyses presented in this paper has duly considered the effect of inspection data on changing both temporal (TU) and parameter (PU) uncertainties of the degradation model. As a result, the multiple-component problem requires a more involved pre-posterior analysis. A question naturally arises: Could this analysis be simplified by ignoring a change in temporal or parameter uncertainty of the gamma process model? This Section quantifies the estimation error caused by such approximations, or naive simplifications of the analysis.

6.1. Ignoring the updating of parameter uncertainty (PU)

Suppose that one uses the inspection data to modify the temporal uncertainty only, and completely neglects the updating of parameter uncertainty.

In this case, the mean value, μ , is treated as a random variable given by the inverse gamma prior, but there is no Bayesian updating given $X(t_1) = x_1$, in the posterior analysis. For example in the single-component problem, the preventive replacement threshold x_{1r} is determined by the same Eq. (22), but with parameters α^{p1} , β^{p1} and ξ_A^{p1} replaced by α , β and ξ_A^π , respectively. With a modified x_{1r} , the pre-posterior analyses given by Eqs. (25) to (27) remains the same. The multiple-component problem is analysed similarly to the single-component problem. Particularly, the posterior failure probability is still given by Eq. (31), but again, the posterior parameters α^{pn} , β^{pn} and ξ_A^{pn} of the scaled F-distribution must be replaced by the same set of prior parameters: α , β and ξ_A^π .

As a result, the ‘posterior’ and ‘pre-posterior’ analyses for the multiple-component problem are exactly the same as the analyses for a single-component problem. Quotation marks are used above because they technically are not the posterior and pre-posterior analyses in the formal sense. Yet they do involve the use of new information in the ‘posterior’ analysis to update the prediction of the failure probability, and the consideration of all possibilities of the information a priori in the pre-posterior analysis. Apparently, these analyses are technically correct only if there is no uncertainty in the mean deterioration rate, i.e., $v_\pi = 0$.

Fig. 8(a) compares the V_1 computed by ignoring the updating of PU with that in the reference case in which changes in both uncertainties are duly incorporated. Fig. 8(b) shows variation of the estimation error with v_π . For a reasonably wide prior with $v_\pi > 0.2$, ignoring

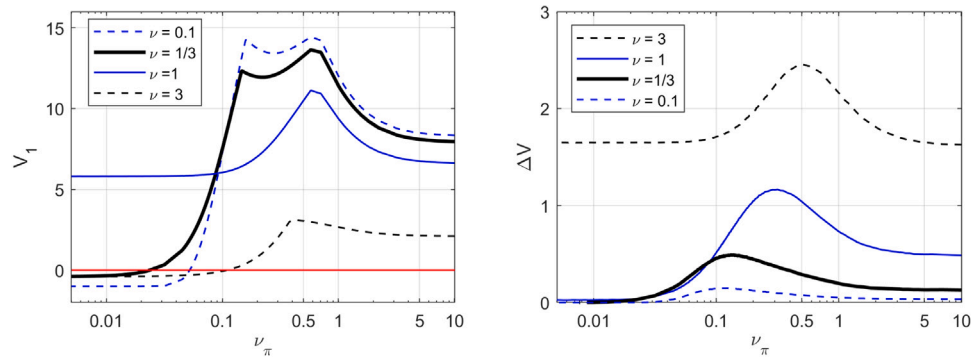


Fig. 7. The effect of ν on the variation of (a) V_1 , and (b) ΔV .

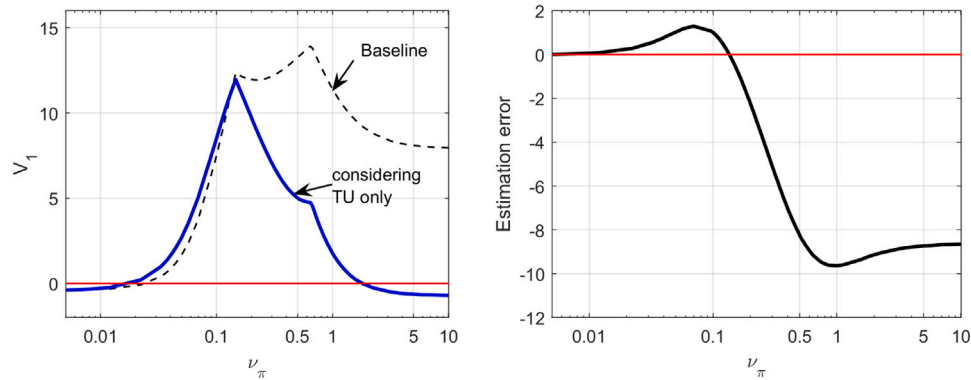


Fig. 8. Effects of ignoring updating of parameter uncertainty: (a) Comparison between V_1 with and without updating of parameter uncertainty; (b) Estimation error.

PU updating results in a large underestimation of V_1 , as evidenced by a negative error in Fig. 8(b). For example, when $\nu_\pi = 0.8$, the maximum estimation error is $-\$9.8$, or -80% of relative error. It is quite remarkable that for a fairly narrow prior, i.e., $\nu_\pi < 0.15$, ignoring PU updating leads to a slight overestimation of V_1 .

6.2. Ignoring the updating of temporal uncertainty (TU)

Given some inspection data, ignoring the updating of TU means performing only the Bayesian updating given $X(t_1) = x_1$. As a result, the posterior failure probability is predicted as if the condition $X(t_1)$ were unknown, although the posterior parameters must be used. Therefore, the same Eq. (9) instead of (15) is used for predicting both the prior and posterior failure probabilities. This underutilization of information was actually observed often in literature, e.g., [34]. Nevertheless, this treatment considerably simplifies the pre-posterior analysis in the multiple-component problem. The reason is that the break-even condition would be completely defined by the sufficient statistics s_n , and the two-dimensional criterion in terms of (x_1, s_{n-1}) will not be required. The computation of a bivariate expectation in Eq. (37) will be simplified to a univariate form similar to Eq. (25).

Results presented in Fig. 9(a) appear to suggest that the error due to ignoring the change in TU is less severe than the error caused by ignoring the reduction in PU. For example, when $\nu_\pi = 0.1$ the maximum estimation error is $-\$4.88$, or -65% of relative error. Of course, these observations are contingent on values of parameters in the reference case, i.e., ν_π , ν , t_1 and t_2 . It is likely that the error due to ignoring the updating of TU would increase or decrease depending on combinations of these parameters.

7. Conclusions

The stochastic gamma process as a model of degradation is widely used in the lifecycle cost analysis literature, where inspection and

maintenance strategies are optimized by minimizing the total expected cost over a system's life cycle. These studies, however, do not account for information generated by inspection itself which can be used to optimize the subsequent maintenance decision. Thus, a question about the value of undertaking an inspection and maintenance program is not answered in a quantitative manner.

To quantify the economic value of an inspection and preventive replacement (I-PR) program, the paper presents the Bayesian pre-posterior analysis involving the gamma process as a model of degradation in a population of components.

The proposed approach uses a conjugate prior, which enables an analytical formulation of the pre-posterior analysis. An original contribution of this study is the refinement of a more familiar Value of Information (VOI) analysis through a careful consideration of the interaction of temporal and parameter uncertainties associated with the gamma process in a multi-component problem. It is interesting to note that the break-even condition in the multiple-component problem is defined by a two-dimensional curve.

The analysis shows that a new piece of information always changes parameter and temporal uncertainties associated with the gamma process model, which is manifested by a shift in the mean and an increase or decrease in the variance of the posterior distribution of future degradation. However, the value is generated only by that information which leads to such an appreciable change in the posterior distribution that also ends up changing the decision as compared to the prior optimal decision without the inspection information.

The proposed model is useful for decision makers who are concerned with investigating ageing management of a component population in a plant that is near its end of operating life.

The numerical, parametric study presented in the paper leads to the following key findings.

- The value of an I-PR program is highly dependent on the prior information about the mean of gamma process, ratio of preventive

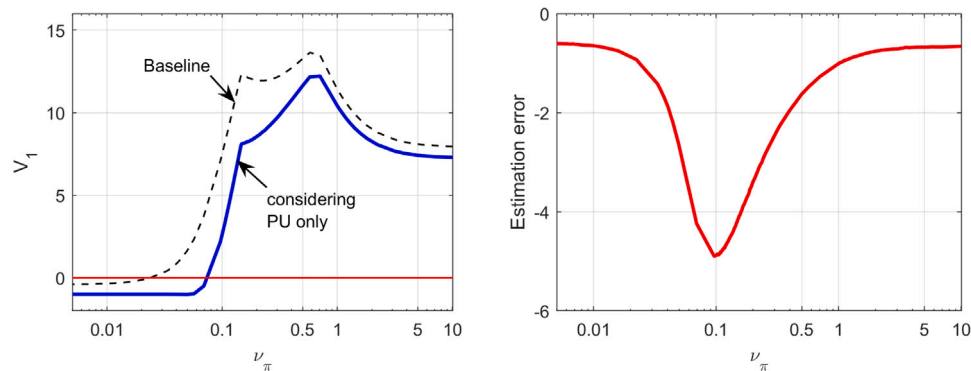


Fig. 9. Effects of ignoring the change in TU for single-component problem: (a) Comparison between V_1 with and without considering the change of TU; (b) Estimation error.

to corrective replacement cost, and variability of the gamma process.

- The value of an I-PR program attains a peak value for a mid-range of dispersion in the prior distribution, i.e., prior COV ranging from 0.1 to 0.5. In case of a diffused prior with COV > 1, the value approaches an asymptotic minimum value.
- The cost ratio has significant effect on the value of an I-PR program. The value is maximum when the ratio of preventive to corrective replacement cost is equal to the prior probability of component failure. This observation appears to be independent of the prior COV.
- In case of a multiple-component population, the inspection data provides an addition value due to synergy of information, which is referred to as the “group effect”. The group effect becomes quite pronounced as the COV of gamma process (i.e., its aleatory variability) increases.
- Simplification of the value analysis by ignoring the change in temporal or parameter uncertainty is not recommended, as it would lead to a large underestimation of the value for a reasonably wide prior distribution (with COV ranging from 0.2 to 1).
- The value of inspection is dominated by a reduction in the parameter uncertainty of the gamma process model. Therefore, an inspection and maintenance optimization problem which ignores the parameter uncertainty is a sub-optimal solution of the problem.

There are a number of potential extensions of this investigation. At present, degradation data are assumed to be free of measurement error. The proposed analysis can be extended to deal with imperfect data resulting from measurement noise and limited detection capability of inspection tools. In addition, this study considers full inspection of the component population. The analysis can be extended to consider the case of partial inspection. However, the calculation of VOI will need to be modified to account for lack of inspection data from a part of component population. Finally, the study uses the gamma process model of degradation, though this can also consider any other stochastic process models, such as the inverse Gaussian processes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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